$$\frac{T_f}{T_i}=\sqrt{\frac{l+\frac{44}{100}l}{l}}=\sqrt{1.44}=1.2$$

Therefore, $T_f = 1.2\ T_i = T_i + 20\%\ T_i$

**Oscillation of liquid in a U-tube:**

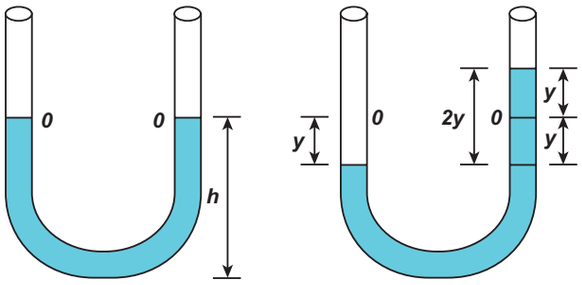



**Figure 10.22** U-shaped glass tube

Consider a U-shaped glass tube which consists of two open arms with uniform cross-sectional area A. Let us pour a non-viscous uniform incompressible liquid of density ρ in the U-shaped tube to a *h*eight *h* as shown in the Figure 10.22. If the liquid and tube are not disturbed then the liquid surface will be in equilibrium position *O*. It means the pressure as measured at any point on the liquid is the same and also at the surface on the arm (edge of the tube on either side), which balances with the atmospheric pressure. Due to this the level of liquid in each arm will be the same. By blowing air one can provide sufficient force in one arm, and the liquid gets disturbed from equilibrium position *O*, which means, the pressure at blown arm is higher than the other arm. This creates difference in pressure which will cause the liquid to oscillate for a very short duration of time about the mean or equilibrium position and finally comes to rest.

Time period of the oscillation is

$$T = 2\pi\sqrt{\frac{l}{2g}}\ \text{second} \qquad (10.62)$$

Where *l* is the total length of liquid column in U- tube.

## 10.5 ENERGY IN SIMPLE HARMONIC MOTION

### a. Expression for Potential Energy

For the simple harmonic motion, the force and the displacement are related by Hooke's law

$$\vec{F} = -k\vec{r}$$

Since force is a vector quantity, in three dimensions it has three components. Further, the force in the above equation is a conservative force field; such a force can be derived from a scalar function which has only one component. In one dimensional case

$$F = -\,k\,x \qquad (10.63)$$

As we have discussed in unit 4 of volume I, the work done by the conservative force field is independent of path. The potential energy U can be calculated from the following expression.

$$F = -\frac{dU}{dx} \qquad (10.64)$$

Comparing (10.63) and (10.64), we get

$$-\frac{dU}{dx} = -\,kx$$

$$dU = k\,x\,dx$$

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**Dummy variable**

The integrating variable $x'$ (read $x'$ as "$x$ prime")is a dummy variable

$$\int_0^y t\,dt = \int_0^y x\,dx = \int_0^y p\,dp = \frac{y^2}{2}$$

Notice that the integrating variables like *t, x* and *p* are dummy variables because, in this integration, whether we put *t* or *x* or *p* as variable for integration, we get the same answer.



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